Transcriptome annotation in the cloud: complexity, best practices and cost

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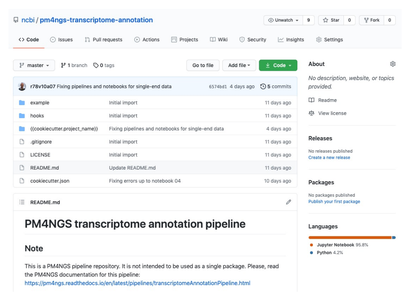
Abstract

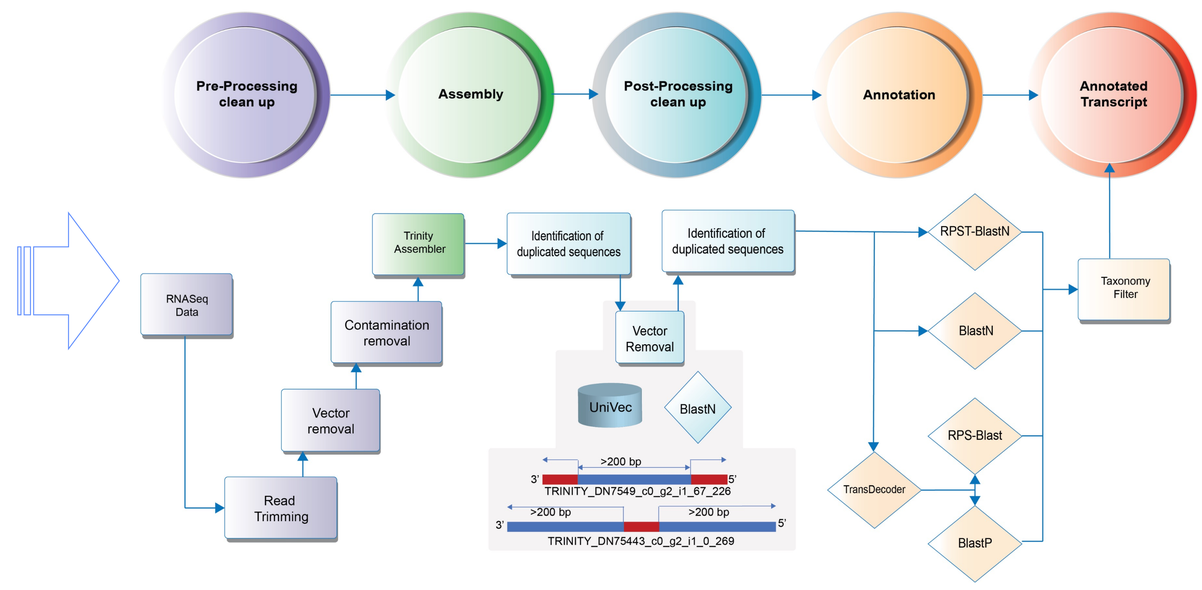
Transcriptome assembly and annotation is a complex process that requires the integration of multiple databases using several computational tools for best results. The annotation process is an important step in developing an understanding of the biological complexity of an organism. The advances in next-generation sequencing technologies and the decrease in the cost of sequencing a complete transcriptome is driving a new era in which annotation is increasingly important and is very productive, especially for the analysis of unannotated organisms. Cloud computing offers a reduced cost of compute resources that makes them accessible for large computational biology experiments such as transcriptome annotation. Here, we describe a complete methodology and pipeline for the assembly and annotation of plant transcriptomes in the Google Cloud Platform (GCP) including two examples for the organisms *Opuntia streptacantha* and *Physalis peruviana*. We demonstrate that public cloud providers are a practical alternative for executing advanced computational biology experiments at relatively low cost. Using our cloud recipes and pipelines, the assembly and annotation of a transcriptome with ~500,000 transcripts can be processed in less than 2 hours with a computing cost of about USD 200-250.

https://pm4ngs.readthedocs.io/en/latest/pipelines/transcriptomeAnnotationPipeline.html

Annotation Workflow

Jupyter Notebooks Templates for PM4NGS





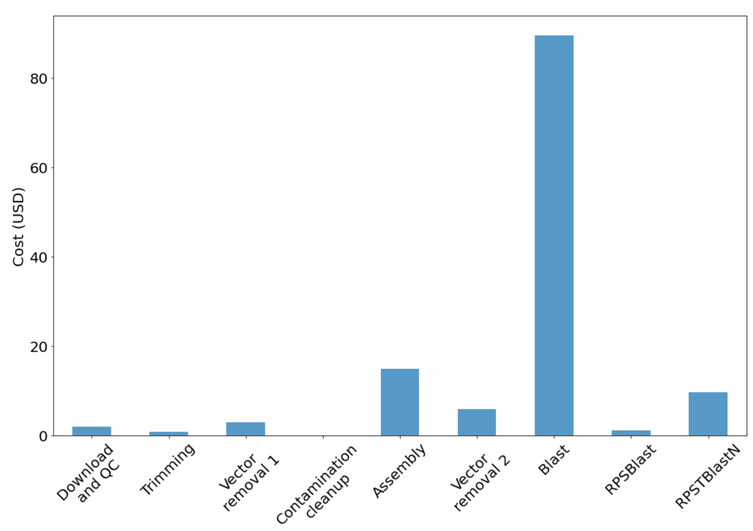
Opuntia streptacantha (BioProject: PRJNA320545)

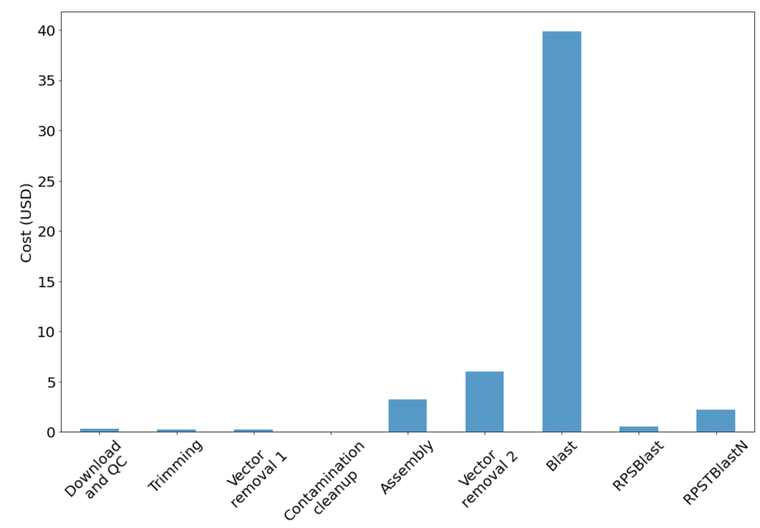
SRA samples: 10
Assembled transcripts: 266,658

Physalis peruviana (BioProject: PRJNA67621)

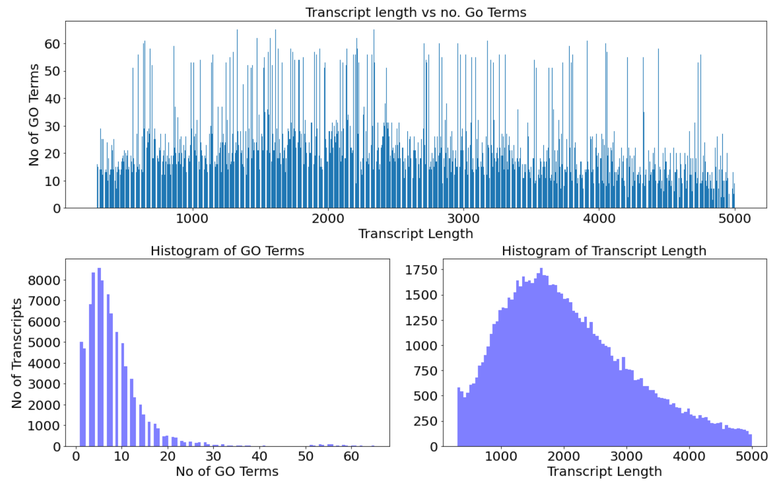
SRA samples: 8
Assembled transcripts: 112,879

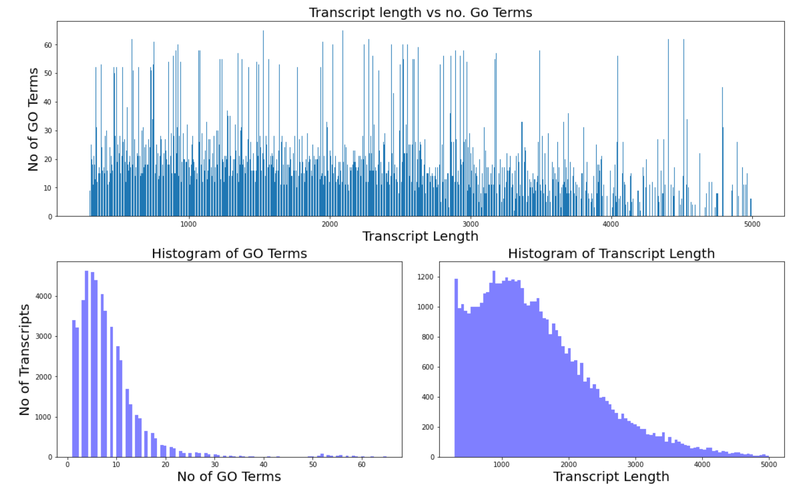
Google Cloud Platform cost



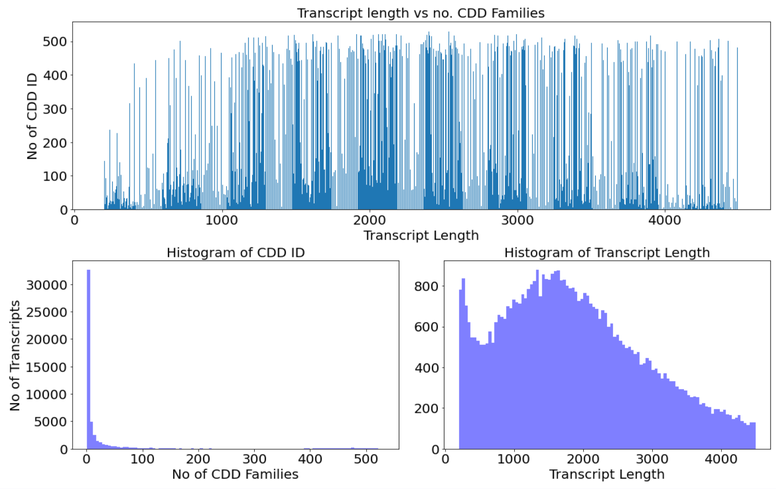


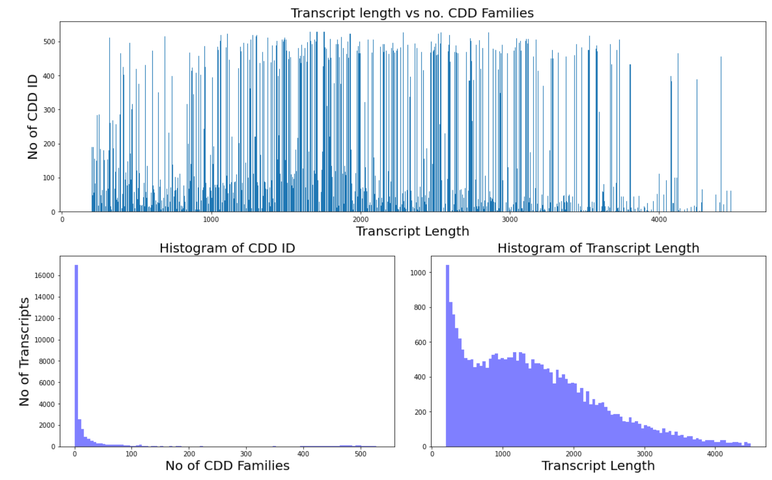
Blast Annotation





CDD Annotation





Discussion

- Transcriptome assembly and annotation pipeline and management using Jupyter notebooks, CWL workflows, and cookiecutter project templates demonstrate the effective of these tools
- Implemented the complete pipeline using the Google Cloud Platform with a compute cost of USD 200-250.
- Data analysis CWL workflows executed from the Jupyter notebook with automatic failing detection and validated with published data
- Transcripts were annotated using gene ontologies (GO) and the conserved domain database (CDD) and submitted to NCBIs Transcriptome Shotgun Assembly Sequence (TSA) Database

Acknowledgment

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References

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3. Vera Alvarez R, Mariño-Ramírez L, Landsman D. Transcriptome annotation in the cloud: complexity, best practices, and cost, *GigaScience*, Volume 10, Issue 2, February 2021. https://pubmed.ncbi.nlm.nih.gov/33511996/